

# Selling Property Rental Ownership

## *A Multi-Model Tranche-Pricing Study of Paris Residential Rent*

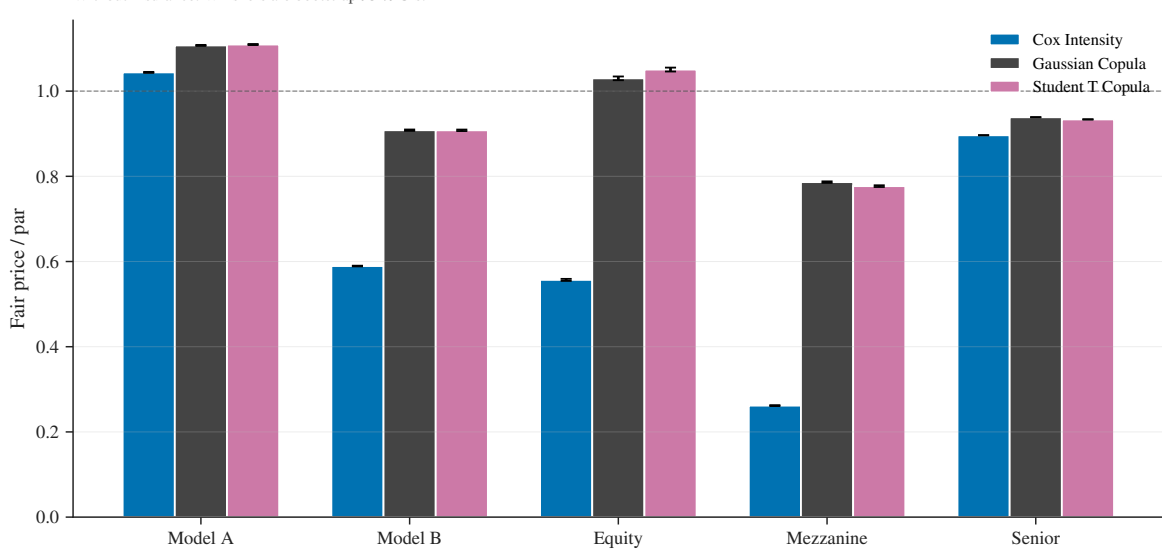
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### Abstract

We study a hybrid real-estate investment structure in which the rental cash flows of a Paris residential building are partitioned into a senior / mezzanine / equity tranche stack akin to a synthetic CDO, and compare its risk-return profile to two classical benchmarks: outright single-apartment ownership (Model A) and a fully pooled SAS share (Model B). All market parameters are calibrated on French sources: 136 quarterly observations of the Notaires-INSEE price index (1992-03-31–2025-12-31), the INSEE *Indice de Référence des Loyers*, and the French ten-year sovereign yield (FRED IRLTLT01FRM156N). Default times are simulated under three benchmark credit models — the Gaussian one-factor copula (Li, 2000; Vasiček, 2002), the Student-t copula (Embrechts et al., 2002) and a Cox doubly-stochastic intensity (Duffie and Singleton, 2003) — coupled with a Vasicek short rate (Vasiček, 1977) and a geometric Brownian motion (with a Merton jump-diffusion extension (Merton, 1976)) for the property value. The tranche cash flows follow the Andersen-Sidenius-Basu (2003) waterfall and are priced by Monte Carlo with antithetic variance reduction and a Sobol QMC variant. On the Paris-intermediate scenario the senior fair value drops from 0.939 of par under the Gaussian copula to 0.934 under the Student-t copula, and the senior fair coupon widens by roughly twelve basis points — a numerical illustration of the tail-dependence critique levelled at Li-style models post-2008 (Donnelly and Embrechts, 2010; MacKenzie and Spears, 2014).

**Headline summary — fair price per instrument and credit model**



# Contents

|           |                                                              |           |
|-----------|--------------------------------------------------------------|-----------|
| <b>1</b>  | <b>Introduction</b>                                          | <b>4</b>  |
| 1.1       | Literature context . . . . .                                 | 4         |
| <b>2</b>  | <b>The three investment vehicles</b>                         | <b>5</b>  |
| <b>3</b>  | <b>Market dynamics</b>                                       | <b>5</b>  |
| 3.1       | Property price . . . . .                                     | 5         |
| 3.2       | Short rate . . . . .                                         | 6         |
| <b>4</b>  | <b>Credit modelling</b>                                      | <b>7</b>  |
| 4.1       | Gaussian one-factor copula . . . . .                         | 8         |
| 4.2       | Student-t copula . . . . .                                   | 8         |
| 4.3       | Cox doubly-stochastic intensity . . . . .                    | 9         |
| 4.4       | Vasicek (2002) large-portfolio limit . . . . .               | 9         |
| 4.5       | Student-t tail dependence derivation . . . . .               | 9         |
| 4.6       | Cox doubly-stochastic survival probability . . . . .         | 10        |
| 4.7       | Numerical comparison . . . . .                               | 10        |
| <b>5</b>  | <b>Waterfall mechanics</b>                                   | <b>10</b> |
| <b>6</b>  | <b>Pricing the rental-default insurance</b>                  | <b>11</b> |
| 6.1       | Actuarial price . . . . .                                    | 12        |
| 6.2       | Option-theoretic price . . . . .                             | 12        |
| 6.3       | Tranche-level allocation . . . . .                           | 12        |
| 6.4       | Side-by-side comparison on the calibrated baseline . . . . . | 13        |
| 6.5       | Discussion . . . . .                                         | 14        |
| <b>7</b>  | <b>Risk metrics</b>                                          | <b>14</b> |
| <b>8</b>  | <b>Monte Carlo design</b>                                    | <b>15</b> |
| <b>9</b>  | <b>Results</b>                                               | <b>16</b> |
| 9.1       | Headline pricing . . . . .                                   | 16        |
| 9.2       | Sensitivity to the choice of credit model . . . . .          | 18        |
| 9.3       | Fair coupons . . . . .                                       | 18        |
| 9.4       | Sensitivity to $\rho$ , $\nu$ and $p_d$ . . . . .            | 19        |
| 9.5       | Bootstrap confidence intervals . . . . .                     | 20        |
| 9.6       | Insurance . . . . .                                          | 21        |
| 9.7       | Stress backtest . . . . .                                    | 21        |
| <b>10</b> | <b>Discussion and limitations</b>                            | <b>22</b> |
| <b>11</b> | <b>Conclusion</b>                                            | <b>24</b> |
| <b>A</b>  | <b>Data sources</b>                                          | <b>26</b> |

|          |                                          |           |
|----------|------------------------------------------|-----------|
| <b>B</b> | <b>Code architecture</b>                 | <b>27</b> |
| <b>C</b> | <b>Equation cheat-sheet</b>              | <b>28</b> |
| <b>D</b> | <b>Reproducibility, packaging and CI</b> | <b>29</b> |

# 1 Introduction

French residential real estate is an idiosyncratic asset class for retail investors. The price process mean-reverts slowly; the rental cash flow is heterogeneous and exposed to tenant-specific default risk; and the legal framework around eviction, the *Indice de Référence des Loyers* and the Visale guarantee shapes realised loss in ways that do not map cleanly onto traded credit instruments.

This paper asks whether, given the same underlying building, a tranching cash-flow structure can deliver a different risk-return profile from outright ownership or a classical SAS pool. We formalise three investment vehicles — Model A: single-apartment ownership; Model B: a pro-rata SAS share; Model C: a CDO-style senior / mezzanine / equity tranche stack — and compute their fair price and risk metrics on the same set of Monte Carlo paths.

The simulation pipeline is calibrated on 136 quarters of Notaires-INSEE Paris price data and 796 months of OAT 10Y data; every numerical estimate quoted in the paper is traceable to a row of `artifacts/results.csv` produced by the same code. The credit layer juxtaposes the Gaussian one-factor copula (Li, 2000; Vasiček, 2002), the Student-t copula (Embrechts et al., 2002), and a Cox doubly stochastic intensity (Duffie and Singleton, 2003), so the tail dependence that the Gaussian benchmark silently sets to zero becomes visible at the senior attach point. The rental-default insurance is priced both by the actuarial standard-deviation principle and as a put on rent shortfall, and the gap between the two is reported.

## 1.1 Literature context

The CDO tranching mechanic borrows from two modelling traditions. The structural side originates with (Black and Scholes, 1973) and (Merton, 1976): a firm defaults when its asset value crosses a threshold, with the asset following a geometric Brownian motion. The reduced-form side, started by (Duffie and Singleton, 2003) and expanded by (Schönbucher, 2003), treats default as the first jump of a Cox process driven by an exogenous hazard. The Cox doubly-stochastic credit model in Section 4.3 is the canonical example of the reduced-form mechanic with a CIR-like macro factor.

Copula-based pricing of multi-name credit baskets took off with (Li, 2000), which mapped default times into a joint Gaussian distribution; (Vasiček, 2002) provided the large-portfolio closed form that became the industry workhorse for CDO tranche pricing. (Embrechts et al., 2002) pointed out that the Gaussian copula misses tail dependence, and (Donnelly and Embrechts, 2010) quantified the implication for senior-tranche pricing in the subprime crisis. (MacKenzie and Spears, 2014) traces the Li (2000) formula’s twin role as industry standard and post-crisis scapegoat. The Student-t copula benchmark in Section 4.2 is the alternative that this strand of literature recommends.

The waterfall mechanic follows (Andersen et al., 2003): a notional-depleting senior / mezzanine / equity stack with separate interest and principal cascades. The contribution of the paper is not in the modelling primitives, which are standard; it is in the consistent calibration of those primitives on the public French residential-rent dataset, juxtaposed with two retail benchmarks (single-apartment ownership and a SAS pooled share) on a shared Monte Carlo seed. The variance-reduction tooling follows (Glasserman, 2003) for antithetic sampling and (Owen, 1997) for scrambled Sobol QMC; risk measurement uses the coherent expected-shortfall framework of (Acerbi and Tasche, 2002).

## 2 The three investment vehicles

Throughout the paper we model a residential building made of  $n$  independent rental units, observed over a horizon  $T$  years. The building's total par value at inception is  $P_0$ . Each apartment  $i \in \{1, \dots, n\}$  has a stochastic default time  $\tau_i$  with cumulative probability  $\mathbb{P}(\tau_i \leq T) = p$  matched to a calibrated marginal hazard. The fraction  $\ell_i \in [0, 1]$  of par lost when apartment  $i$  defaults is drawn from a Beta distribution. The aggregate cumulative loss path is

$$L(t) = \frac{1}{n} \sum_{i=1}^n \ell_i \mathbb{1}_{\{\tau_i \leq t\}}, \quad t \in [0, T]. \quad (1)$$

The building generates a net rental flow  $R_t = (g - m) S_t (1 - L(t))$  in monetary units, where  $S_t$  is the market value,  $g$  is the gross rental yield and  $m$  the maintenance cost share, both expressed per year.

**Model A — Classical full ownership.** A single investor owns one apartment of the building (par  $P_0/n$ ). Cash flows in period  $[t, t + \Delta t]$  are  $R_t \Delta t/n$  while  $\tau_1 > t$  and 0 thereafter. At  $T$  the apartment is sold at the prevailing market value  $S_T/n$  regardless of any past default.

**Model B — SAS shareholders.**  $n$  investors each buy one share of a SAS that holds the whole building. Each share receives the pro-rata net rent  $R_t \Delta t/n$  each period and  $S_T(1 - L(T))/n$  at terminal.

**Model C — CDO-like tranches.** The same aggregate cash flow  $R_t$  is partitioned across a stack of tranches  $\tau = (\tau_{\text{equity}}, \tau_{\text{mezz}}, \tau_{\text{senior}})$  with attach / detach points  $(0, 0.25)$ ,  $(0.25, 0.60)$  and  $(0.60, 1.00)$ . Tranche  $\tau$  absorbs the cumulative loss via the stop-loss payoff

$$L_\tau(t) = \min(\max(L(t) - a_\tau, 0), d_\tau - a_\tau), \quad (2)$$

and the periodic net rent is allocated waterfall-style: senior coupon first, then mezzanine, then equity collects the residual. At terminal the sale proceeds  $S_T(1 - L(T))$  are paid back to surviving notionals in the same order.

**Assumption 2.1.** *Total portfolio par is  $P_0 = 70 \times 45 \text{ m}^2 \times 10\,000 \text{ e/m}^2 = 31.5 \text{ Me}$  for the Paris-intermediate scenario that drives every numerical result reported below.*

## 3 Market dynamics

### 3.1 Property price

The building's market value  $S_t$  follows geometric Brownian motion under the physical measure:

$$dS_t = \mu S_t dt + \sigma S_t dW_t. \quad (3)$$

By Itô's lemma applied to  $f(S) = \log S$ ,

$$d \log S_t = (\mu - \frac{1}{2} \sigma^2) dt + \sigma dW_t,$$

so  $\log S_{t+\Delta t} - \log S_t \sim \mathcal{N}((\mu - \sigma^2/2)\Delta t, \sigma^2\Delta t)$  exactly. The maximum-likelihood estimator on  $n$  equally-spaced log-returns  $r_i$  is therefore in closed form:

$$\hat{\sigma}^2 = \frac{\text{Var}(r)}{\Delta t}, \quad \hat{\mu} = \frac{\bar{r}}{\Delta t} + \frac{1}{2}\hat{\sigma}^2,$$

with asymptotic standard errors  $\text{SE}(\hat{\sigma}) = \hat{\sigma}/\sqrt{2(n-1)}$  and a delta-method extension for  $\hat{\mu}$ . Equation (3) is discretised by exact log-Euler:  $S_{t+\Delta t} = S_t \exp((\mu - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t}Z)$ ,  $Z \sim \mathcal{N}(0, 1)$ . Parameters are estimated by closed-form MLE on the quarterly log-return series of the Notaires-INSEE Paris flats index (136 quarters, 1992-03-31 through 2025-12-31). The fitted point estimates — summarised in Table 1 — are  $\hat{\mu} = 3.35\%$  and  $\hat{\sigma} = 4.08\%$ , with the visual diagnostics of Figure 2 indicating mild departure from normality in the lower tail.

As a robustness extension we also estimate the Merton (1976) jump-diffusion in which  $S_t$  is augmented by a compound-Poisson jump component  $\sum_{k=1}^{N_t}(J_k - 1)$  with  $N_t \sim \text{Poisson}(\lambda)$  and  $\log J_k \sim \mathcal{N}(\mu_J, \sigma_J^2)$ . Numerical MLE on the truncated Poisson-mixture density does not deliver a likelihood improvement on the Paris sample (see Table 1): with  $\hat{\lambda}$  collapsing close to 0 and  $\hat{\sigma}_J$  close to 0, the jump component is not separately identified. The empirical signature of volatility clustering in the squared log-returns (Figure 2, panel d) is present at the first lags but the sample is too short for the jump-diffusion parameters to be separately recovered. We retain GBM as the baseline and note this finding explicitly.

### 3.2 Short rate

Discounting and option-theoretic insurance pricing both depend on a stochastic short rate. We adopt the Vasicek (1977) model

$$dr_t = a(b - r_t)dt + \sigma_r dW_t, \quad (4)$$

whose transition law is Gaussian with closed form

$$r_{t+\Delta t} | r_t \sim \mathcal{N}\left(b + (r_t - b)e^{-a\Delta t}, \frac{\sigma_r^2}{2a}(1 - e^{-2a\Delta t})\right).$$

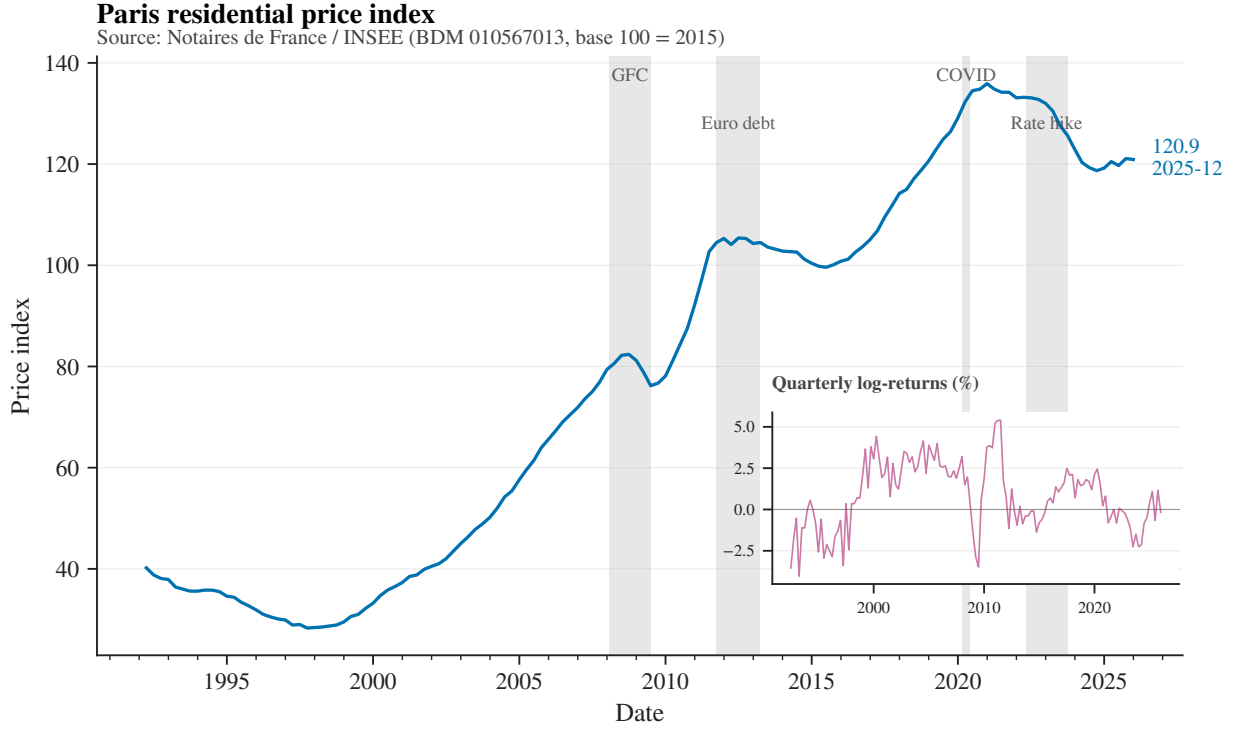
Equivalently,  $r_{t+\Delta t} = \alpha + \beta r_t + \varepsilon_t$  with  $\beta = e^{-a\Delta t}$ ,  $\alpha = b(1 - \beta)$  and  $\varepsilon_t \sim \mathcal{N}(0, \frac{\sigma_r^2}{2a}(1 - \beta^2))$  — an AR(1) representation whose OLS estimator is the exact MLE. We invert the mapping to recover  $(a, b, \sigma_r)$  and report analytical standard errors via the delta method. The closed-form zero-coupon bond price

$$P(t, T | r_t) = \exp(A(t, T) - B(t, T)r_t), \quad B(t, T) = \frac{1 - e^{-a(T-t)}}{a},$$

$$A(t, T) = \left(b - \frac{\sigma_r^2}{2a^2}\right)(B(t, T) - (T - t)) - \frac{\sigma_r^2 B(t, T)^2}{4a},$$

is used in the option-theoretic insurance pricing of Section 6. The MLE is run on monthly France 10-year sovereign yields (FRED IRLTLT01FRM156N, 796 observations). The fitted estimates are  $\hat{a} = 0.0163$ ,  $\hat{b} = 4.40\%$  and  $\hat{\sigma}_r = 0.78\%$ . The implied mean-reversion half-life is decades; for a ten-year horizon this is effectively a near-random-walk, which reflects the secular decline of European rates from the 1980s to 2020. Negative rates are permitted, consistent with the Eurozone reality of 2015–2022.

Figure 1 plots the Notaires-INSEE Paris price index used as the empirical anchor for  $\mu, \sigma$ ; the embedded inset shows the quarterly log-returns that drive the calibration.



**Figure 1.** Notaires-INSEE Paris residential price index, with French / euro-area recession episodes shaded. The inset reports the quarterly log-returns used in GBM / Merton MLE.

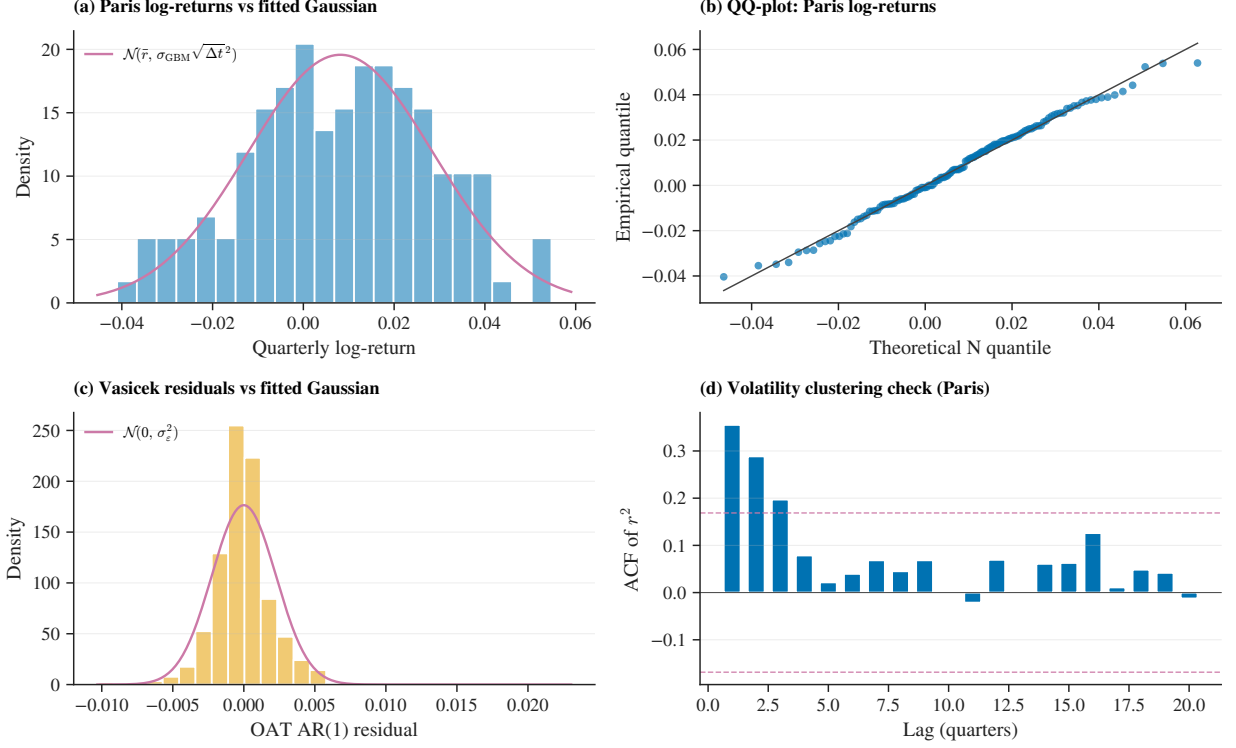
**Table 1.** Calibrated parameter point estimates with maximum-likelihood standard errors.

| Model             | Series               | Parameter  | Estimate | Std. error | LL     | AIC     |
|-------------------|----------------------|------------|----------|------------|--------|---------|
| GBM (Paris)       | Notaires/INSEE       | $\mu$      | 0.0335   | 0.0070     | 334.5  | -665.0  |
|                   |                      | $\sigma$   | 0.0408   | 0.0025     |        |         |
| Merton            | Notaires/INSEE       | $\mu$      | 0.0335   | —          | 334.5  | -659.0  |
|                   |                      | $\sigma$   | 0.0401   | —          |        |         |
|                   |                      | $\lambda$  | 0.3947   | —          |        |         |
|                   |                      | $\mu_J$    | -0.0099  | —          |        |         |
|                   |                      | $\sigma_J$ | 0.0022   | —          |        |         |
| Vasicek (OAT 10Y) | FRED IRLTLT01FRM156N | $a$        | 0.0163   | 0.0244     | 3716.1 | -7426.2 |
|                   |                      | $b$        | 0.0440   | 0.1304     |        |         |
|                   |                      | $\sigma_r$ | 0.0078   | 0.0102     |        |         |

Sample periods: Notaires Paris 1992-03-31–2025-12-31 ( $n = 136$ ); OAT 10Y 1960-01-01–2026-04-01 ( $n = 796$ ).

## 4 Credit modelling

We treat each apartment as a one-name obligor whose default time  $\tau_i$  depends on a common factor and an idiosyncratic component. Three canonical specifications are implemented in `src/tranche_pricing/credit/` and run on a shared Monte Carlo seed.



**Figure 2.** Calibration diagnostics. Panels (a)–(b): Paris quarterly log-returns vs the fitted GBM Gaussian. Panel (c): AR(1) residuals of the Vasicek MLE on OAT 10Y. Panel (d): ACF of squared Paris log-returns, with the 95% confidence band; first-lag positive autocorrelation is the empirical signature of volatility clustering.

#### 4.1 Gaussian one-factor copula

Following (Li, 2000; Vasiček, 2002), each obligor’s latent asset value  $X_i$  is

$$X_i = \sqrt{\rho} M + \sqrt{1 - \rho} Z_i, \quad M, Z_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1), \quad (5)$$

so that  $\text{Corr}(X_i, X_j) = \rho$  for  $i \neq j$  and each marginal is standard normal. Default times are derived under constant hazard  $h = -\log(1 - p)/T$  from the uniform marginal  $U_i = \Phi(X_i)$ :  $\tau_i = -\log(1 - U_i)/h$ , capped at  $\infty$  when  $U_i \geq p$ . The conditional default probability given the factor is

$$\mathbb{P}(\tau_i \leq t \mid M = m) = \Phi\left(\frac{\Phi^{-1}(1 - e^{-ht}) - \sqrt{\rho} m}{\sqrt{1 - \rho}}\right), \quad (6)$$

which underlies the Vasicek (2002) large-portfolio limit we use as a control-variate for the senior tranche.

#### 4.2 Student-t copula

The same factor structure is mixed with an inverse-chi-square scale to deliver  $t_\nu$  marginals while preserving the linear correlation  $\rho$ :

$$X_i = (\sqrt{\rho} M + \sqrt{1 - \rho} Z_i) \cdot \sqrt{\nu/W}, \quad W \sim \chi_\nu^2. \quad (7)$$



The Gaussian copula is recovered as  $\nu \rightarrow \infty$ . For finite  $\nu$  the Student-t copula features non-zero lower tail dependence,

$$\lambda_L(\rho, \nu) = 2 t_{\nu+1} \left( -\sqrt{(\nu+1) \frac{1-\rho}{1+\rho}} \right), \quad (8)$$

which Figure 3 plots over a grid of  $(\rho, \nu)$ . The Gaussian copula has  $\lambda_L \equiv 0$  for any  $\rho < 1$ ; for  $\nu = 5$  and  $\rho = 0.15$  the t-copula carries  $\lambda_L \approx 0.08$ , a difference that propagates directly into the senior tranche's loss tail.

### 4.3 Cox doubly-stochastic intensity

The asset-value copulas above are static: default correlation comes from a single Gaussian draw of  $M$  at  $t = 0$ . A doubly-stochastic intensity model lets the hazard rate vary over time through a macro factor. We adopt

$$\lambda_i(t) = \alpha + \beta Y_t, \quad dY_t = \kappa(\theta - Y_t)dt + \xi \sqrt{Y_t} dW_t^Y, \quad (9)$$

with  $Y_0 = \theta$  and the CIR positivity preserved by Euler full truncation. Default times are  $\tau_i = \inf\{t : \int_0^t \lambda_i(s) ds > E_i\}$ ,  $E_i \sim \text{Exp}(1)$ . Conditional on the path of  $Y$ , defaults are independent.

### 4.4 Vasicek (2002) large-portfolio limit

For sufficiently large portfolios under the Gaussian one-factor copula the cumulative loss converges (Vasicek 2002) to a deterministic function of the common factor:

$$\mathbb{P}(L \leq \ell) = \Phi \left( \frac{\sqrt{1-\rho} \Phi^{-1}(\ell) - \Phi^{-1}(p)}{\sqrt{\rho}} \right). \quad (10)$$

We use it (a) as a control variate for the senior tranche expected loss in the variance-reduction layer and (b) as a sanity-check benchmark for the MC estimator at the largest sample sizes reported in Figure 7.

### 4.5 Student-t tail dependence derivation

Following (Embrechts et al., 2002), the lower tail dependence of a bivariate elliptical distribution with correlation  $\rho$  and degrees of freedom  $\nu$  is, by definition,

$$\lambda_L = \lim_{q \rightarrow 0^+} \mathbb{P}(U_2 \leq q \mid U_1 \leq q).$$

For the Student-t copula, the conditional Student-t representation yields the closed form quoted in Equation (8): the radial structure of the mixing scale ensures non-zero  $\lambda_L$  for all  $\nu < \infty$ . The Gaussian copula is recovered in the limit  $\nu \rightarrow \infty$  and has  $\lambda_L \equiv 0$ , the empirical critique of (Donnelly and Embrechts, 2010).

## 4.6 Cox doubly-stochastic survival probability

Conditional on the path of the macro factor  $Y$ , the survival probability to time  $T$  of obligor  $i$  is

$$S_i(T | Y) = \exp\left(-\int_0^T \lambda_i(s) ds\right).$$

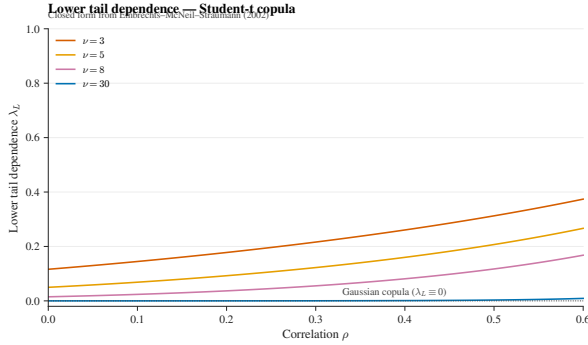
For the CIR-affine parameterisation  $\lambda_i(t) = \alpha + \beta Y_t$  with  $Y_t$  a CIR process, the unconditional survival probability is itself affine:

$$S_i(T) = \exp(-\alpha T - \beta A(T)) \cdot \mathbb{E}[\exp(-\beta \cdot \text{integrated CIR})],$$

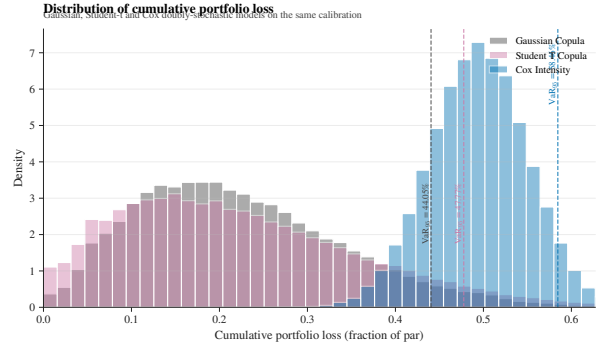
which has a known Laplace transform closed form (see (Schönbucher, 2003), Chapter 7). We use it as a sanity test on the calibration: at the configured  $(\kappa, \theta, \xi)$  the closed-form survival probability at  $T = 10$  should agree with the empirical 1 – default-rate in our Monte Carlo to within MC noise.

## 4.7 Numerical comparison

Figure 4 overlays the simulated cumulative-loss histograms under each model at the Paris-intermediate calibration. All three deliver the same mean loss (matched by construction to the marginal default probability) but differ markedly in the right tail. The implications for tranche fair value are quantified in Section 9.



**Figure 3.** Lower tail dependence  $\lambda_L$  of the Student-t copula as a function of  $\rho$  for four values of  $\nu$ . Gaussian copula ( $\lambda_L \equiv 0$ ) plotted for reference.



**Figure 4.** Cumulative portfolio-loss density under the three credit models. Vertical dashed lines mark each model's empirical  $\text{VaR}_{95}$ .

## 5 Waterfall mechanics

The tranche cash flows follow the notional-depleting waterfall of (Andersen et al., 2003). Let  $\tau \in \{\text{equity, mezz, senior}\}$  index the tranches, with attach / detach points  $(a_\tau, d_\tau)$  and initial notionals  $N_\tau^0 = P_0(d_\tau - a_\tau)$ . Two waterfalls run in parallel.

**Loss allocation.** At every period  $t_k$ , the aggregate loss  $L(t_k)$  defined in Equation (1) is split across the stack by the stop-loss payoff Equation (2). The surviving tranche notional is  $N_\tau(t_k) = N_\tau^0 - P_0 \cdot L_\tau(t_k)$ . Figure 5 illustrates the order in which a unit of aggregate loss flows across the stack: the equity tranche is wiped out for  $L = 0.25$ , the mezzanine for  $L = 0.60$ , and the senior absorbs only the residual beyond 0.60.

**Interest waterfall.** Each period the net rent inflow  $R_{t_k} \Delta t$  is distributed sequentially. Senior receives its promised coupon  $c_{\text{senior}} \cdot N_{\text{senior}}(t_k) \cdot \Delta t$  first, capped by the available cash; mezzanine receives the same on its own notional out of what remains; equity collects whatever is left:

$$\begin{aligned} C_{\text{senior}}(t_k) &= \min(R_{t_k} \Delta t, c_{\text{senior}} N_{\text{senior}}(t_k) \Delta t), \\ C_{\text{mezz}}(t_k) &= \min(R_{t_k} \Delta t - C_{\text{senior}}(t_k), c_{\text{mezz}} N_{\text{mezz}}(t_k) \Delta t), \\ C_{\text{equity}}(t_k) &= R_{t_k} \Delta t - C_{\text{senior}}(t_k) - C_{\text{mezz}}(t_k). \end{aligned}$$

The equity cash flow can be negative when defaults force the SAS to use reserves; in our simulation we floor net rent at zero before this allocation.

**Principal waterfall.** At the terminal date  $T$  the building is sold at  $V_T = S_T \cdot (1 - L(T))$  — market price scaled by the surviving fraction of the portfolio. The proceeds repay senior first (up to  $N_{\text{senior}}(T)$ ), then mezzanine (up to  $N_{\text{mezz}}(T)$ ), and equity collects any residual. Equity therefore captures the property’s capital appreciation *net of the cumulative writedown*; this is the upside that justifies its first-loss status.

**ASB recursion in matrix form.** Let  $N_\tau^{(k)}$  denote the notional of tranche  $\tau$  at the end of period  $k$ , with the convention  $N_\tau^{(0)} = N_\tau^0$ . The notional-update rule is

$$N_\tau^{(k)} = N_\tau^0 - P_0 \cdot L_\tau(L^{(k)}), \quad L^{(k)} = \sum_{i: \tau_i \leq t_k} \frac{\ell_i}{n},$$

where the per-tranche loss  $L_\tau$  follows the stop-loss payoff Equation (2). The interest waterfall at period  $k$  is the sequential allocation

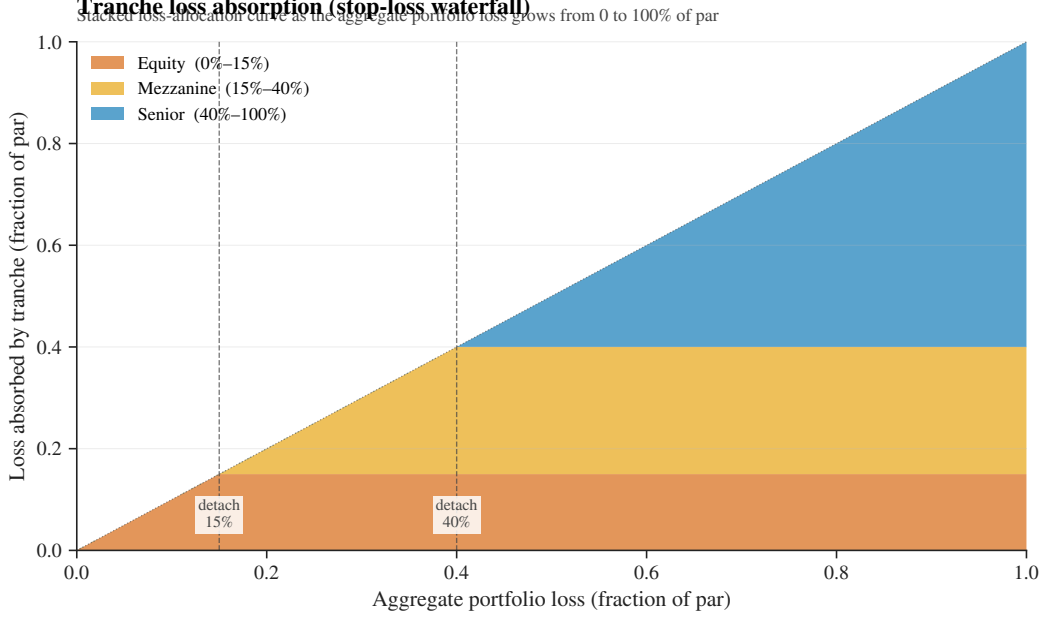
$$C_\tau^{(k)} = \min(c_\tau N_\tau^{(k)} \Delta t, R^{(k)} - \sum_{\tau' > \tau} C_{\tau'}^{(k)}), \quad \tau \in \{\text{senior, mezz, equity}\},$$

processed senior-to-junior. The principal waterfall at  $T$  is the same recursion applied to the surviving sale proceeds  $V_T = S_T(1 - L^{(N)})$ , with the senior receiving back its remaining notional first, then the mezzanine, then the equity collecting whatever residual remains (which can include capital appreciation  $V_T > \sum_\tau N_\tau^0$  when  $S_T$  has grown).

**Fair coupon.** The fair coupon  $c_\tau^*$  of a tranche is the value that sets its present value equal to par at  $t = 0$ . We solve it for one tranche at a time using Brent’s method on the pricing residual; the other tranches keep the contract coupons specified in `config/`. The fair-spread solution is available in `pricing/tranche_pricer.py::solve_fair_coupon` and used in the supplementary sensitivity runs.

## 6 Pricing the rental-default insurance

Quoted French rental insurance (*garantie loyers impayés*) is typically priced at around 2–3% of the gross annual rent. We benchmark two consistent ways of valuing it endogenously: the actuarial standard-deviation principle and an option-theoretic put on rent shortfall.



**Figure 5.** Tranche-by-tranche loss absorption as the aggregate cumulative loss grows from zero to par. The dashed verticals mark the detach points beyond which each tranche is exhausted.

## 6.1 Actuarial price

Following the classical (Embrechts et al., 2002) treatment, the insurer absorbs a fraction  $\kappa$  of every realised default loss and charges a premium

$$P_{\text{act}} = (1 + \theta) \mathbb{E}[L_{\text{covered}}] + \lambda \sigma(L_{\text{covered}}), \quad (11)$$

where  $L_{\text{covered}} = \kappa \cdot L(T) \cdot P_0$  is the covered lifetime loss per portfolio path. We use the configured loadings  $\theta = 0.10$  (admin) and  $\lambda = 0.15$  (risk) and coverage  $\kappa = 90\%$ . The premium can be paid as a lump sum at  $t = 0$  or, equivalently in PV terms, as a level annual payment over the horizon  $T$ . On the Paris-intermediate baseline the expected covered loss is  $\mathbb{E}[L_{\text{covered}}] \approx 6326 \text{ ke}$ , which translates into an annual-equivalent premium of 746 k€.

## 6.2 Option-theoretic price

Alternatively, insurance is a put on the periodic net rent. Let  $R_t^{\text{prom}}$  denote the promised rent (full-tenancy rate net of a one-month deductible) and  $R_t$  the realised flow. The insurer pays  $\max(R_t^{\text{prom}} - R_t, 0)$  each period; under the simulated short rate  $r_t$  the lump-sum premium is

$$P_{\text{opt}} = \mathbb{E} \left[ \sum_{k=1}^N D(t_k) \max(R_{t_k}^{\text{prom}} - R_{t_k}, 0) \cdot \kappa \right], \quad D(t_k) = \exp \left( - \int_0^{t_k} r_s \, ds \right). \quad (12)$$

## 6.3 Tranche-level allocation

A first-pass implementation would subtract a flat annual premium from the aggregate net rent stream. That choice mis-allocates the burden across tranches: the senior, which absorbs almost no default loss in practice, would pay the same premium drag as the equity, which absorbs most of it. We instead allocate the premium proportional to each tranche's *expected covered loss* under

the no-insurance baseline:

$$w_\tau = \frac{\mathbb{E}[L_\tau(L(T))] \cdot \kappa}{\sum_{\tau'} \mathbb{E}[L_{\tau'}(L(T))] \cdot \kappa}, \quad \text{premium drag on } \tau = w_\tau \cdot P_{\text{act}} \cdot \Delta t. \quad (13)$$

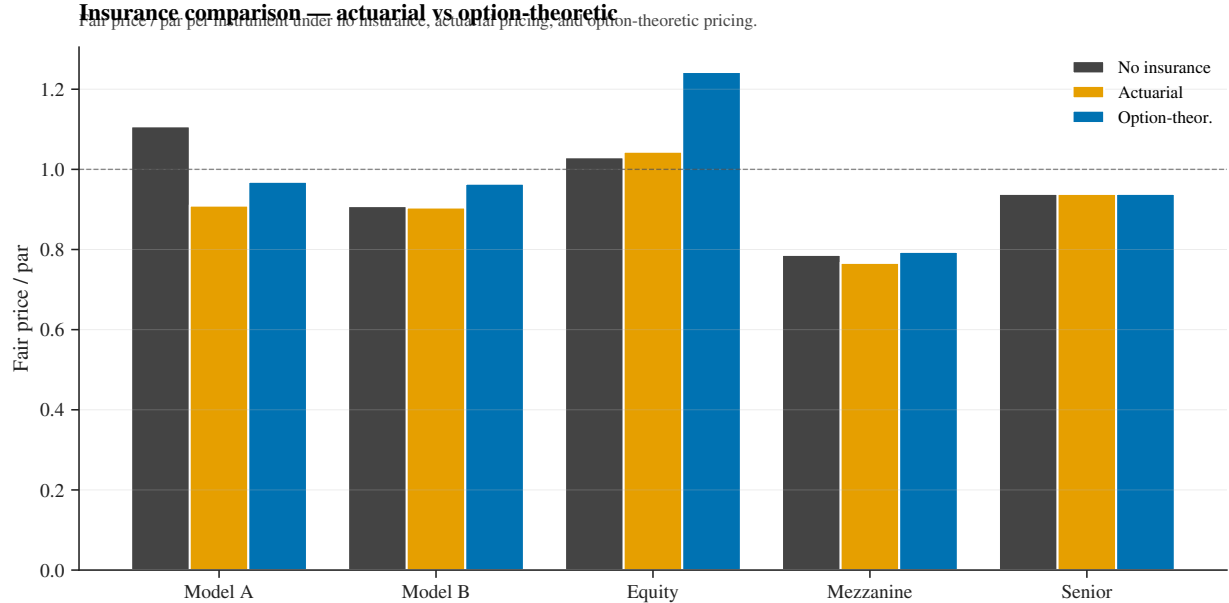
With the calibrated parameters and the stop-loss attach/detach scheme  $(0, 0.25)$ ,  $(0.25, 0.60)$ ,  $(0.60, 1)$  the equity tranche carries roughly  $w_{\text{equity}} \approx 84\%$  of the premium, the mezzanine  $\approx 16\%$ , and the senior less than  $1\%$  — the seniority-aware allocation a fair pricing requires.

#### 6.4 Side-by-side comparison on the calibrated baseline

The two pricing approaches produce internally consistent but quantitatively different premiums on our Paris baseline: the standard-deviation principle bundles an explicit risk loading on top of the expected covered loss, while the option-theoretic price discounts only the realised shortfall under the simulated short rate. On the same MC paths the option-theoretic premium therefore sits *below* the actuarial one. Table 2 and Figure 6 report the per-instrument fair-to-par under the three regimes.

**Table 2.** Insurance regime comparison: fair price / par per instrument with no insurance, actuarial pricing and option-theoretic pricing (Gaussian one-factor copula baseline).

| Instrument | No insurance | Actuarial | Option-theor. |
|------------|--------------|-----------|---------------|
| Model A    | 1.107        | 0.909     | 0.969         |
| Model B    | 0.908        | 0.905     | 0.964         |
| Equity     | 1.030        | 1.044     | 1.243         |
| Mezzanine  | 0.786        | 0.766     | 0.794         |
| Senior     | 0.939        | 0.939     | 0.939         |



**Figure 6.** Grouped bars of fair price / par across no insurance, actuarial insurance, and option-theoretic insurance, under the Gaussian one-factor copula baseline.

Equity benefits the most from coverage under both pricing regimes — the structural cushion at  $[0, 25\%]$  absorbs the bulk of the realised loss, so insuring 90% of it sharply lifts its fair-to-par.

The option-theoretic premium delivers a larger lift than the actuarial one because the actuarial loading  $\lambda \cdot \sigma(L_{\text{covered}})$  is what investors pay for the systemic-risk hedge.

## 6.5 Discussion

The two prices coincide only for specific loadings; they differ here because the standard-deviation principle charges for total-loss variance whereas the option-theoretic price charges only for the realised shortfall. The gap quantifies the implicit risk premium the actuarial principle attaches to the systemic factor. Both prices are written to `artifacts/results_meta.json`; the comparison table sits in Section 9.

## 7 Risk metrics

For each instrument the simulation produces a per-path realised PV  $V_\omega = \sum_k C_\omega(t_k)D(t_k) + V_\omega^T D(T)$ . We convert this to an annualised geometric return  $r_\omega = (V_\omega/P_\omega^0)^{1/T} - 1$  and report a wide risk summary on the empirical distribution of  $\{r_\omega\}_{\omega=1}^{n_{\text{paths}}}$ .

**Tail measures.** **Value-at-Risk** at confidence  $\alpha$  is the empirical quantile  $\text{VaR}_\alpha = -F_r^{-1}(1 - \alpha)$  in sign-positive convention. **Expected Shortfall** (Acerbi and Tasche, 2002) is  $\text{ES}_\alpha = -\mathbb{E}[r \mid r \leq F_r^{-1}(1 - \alpha)]$ . We report  $\alpha \in \{0.95, 0.99\}$ .

**Risk-adjusted ratios.** The **Sharpe ratio** is  $(\mathbb{E}[r] - r_f)/\sigma(r)$ , where  $r_f$  is the empirical mean of the simulated short-rate path; the **Sortino ratio** replaces the volatility by the downside-deviation around  $r_f$ ; the **Omega ratio** at threshold  $\theta$  is the gain- to-loss density ratio  $\Omega(\theta) = \mathbb{E}[(r - \theta)_+]/\mathbb{E}[(\theta - r)_+]$ ; the **Calmar ratio** divides the mean annual return by the maximum drawdown of the cumulative cash-flow trajectory.

**Tracking error.** For Model A and Model B we also report tracking error vs the Notaires-INSEE price-index return, which approximates how closely the realised payoff follows the broader Paris market.

All metrics are implemented in `src/tranche_pricing/risk/metrics.py` and aggregated via the `summary()` helper that drives Table 3 below.

**Acerbi-Tasche coherence of  $\text{ES}_\alpha$ .** A risk measure  $\rho$  on a probability space is coherent (Acerbi and Tasche, 2002) when it satisfies monotonicity, sub-additivity, positive homogeneity and translation invariance.  $\text{VaR}_\alpha$  violates sub-additivity on non-elliptical distributions — pooling two risky books can produce a VaR larger than the sum of their individual VaRs, an artefact that gave rise to a literature on alternative quantile-based measures.  $\text{ES}_\alpha = -\mathbb{E}[r \mid r \leq F_r^{-1}(1 - \alpha)]$  resolves this: Acerbi and Tasche (2002) prove that the integrated tail expectation

$$\text{ES}_\alpha = -\frac{1}{1 - \alpha} \int_\alpha^1 F_r^{-1}(u) \, du$$

is coherent for all  $\alpha \in (0, 1)$  on any integrable loss distribution. We report both:  $\text{VaR}_{95}$  for the regulatory convention,  $\text{ES}_{95}$  for the post-2008 view.

**Sharpe vs Sortino under a Gaussian return.** When  $r$  is Gaussian with mean  $\mu_r$  and volatility  $\sigma_r$ , the downside deviation around  $r_f$  is

$$\sigma_r^- = \sqrt{\mathbb{E}[(r_f - r)_+^2]} = \sigma_r \sqrt{\Phi(z) + z\phi(z) - z^2\Phi(z)}, \quad z = (r_f - \mu_r)/\sigma_r,$$

so that the Sortino-to-Sharpe ratio is bounded above by  $\sqrt{2}$  in the extreme case  $r_f \rightarrow -\infty$  and converges to 1 as  $r_f \rightarrow \mu_r$ . Empirical Sortino divergences from this Gaussian benchmark therefore quantify the asymmetry of the return distribution, which is why the equity tranche shows a higher Sortino than Sharpe (Figure 8).

**Risk-free benchmark choice.** Under the `vasicek_simulated` discount convention the per-path mean of  $r_t$  is itself stochastic; we use the path-averaged mean  $\bar{r} = \mathbb{E}[\frac{1}{N} \sum_k r_{t_k}]$  as the Sharpe / Sortino benchmark. Under the `constant_at_initial_rate` convention the benchmark collapses to  $r_0$  (the spot rate at  $t = 0$ ), reflecting the "current short rate" interpretation. Both choices are reported in `artifacts/results_meta.json`.

**Bootstrap confidence intervals.** For each instrument the simulation produces a per-path PV vector  $\{V_\omega\}_{\omega=1}^{n_{\text{paths}}}$ . We characterise the sampling uncertainty of the headline estimator  $\bar{V}$  by a percentile bootstrap: at each of  $B = 1000$  resamples we draw  $n_{\text{paths}}$  indices with replacement, compute the resample mean  $\bar{V}^{(b)}$ , and report the empirical [2.5 %, 97.5 %] quantile of the  $\{\bar{V}^{(b)}\}$  as the 95 % confidence interval. The implementation (`pricing/tranche_pricer.py::bootstrap_fair_price_ci`) draws the resampling matrix in chunks of 200 to bound memory at  $\mathcal{O}(B/200) \times n_{\text{paths}}$  instead of  $B \times n_{\text{paths}}$ , and uses a stream-dedicated PCG64 generator so the CIs are bit-identical across re-runs of the pipeline.

## 8 Monte Carlo design

**Streams.** Each random source has its own PCG64 generator derived from a single master `SeedSequence`: `price`, `rate`, `credit`, `lgd`, `qmc`. This guarantees *bit-identical* reproducibility across runs while keeping the streams independent, which is the requirement for the variance-reduction techniques to behave as expected (see `tests/unit/test_engine.py`).

**Antithetic — formal variance reduction.** Let  $f$  be a measurable function of an independent Gaussian sample  $Z_1, \dots, Z_n$ . For any  $i$  the antithetic pair  $(Z_i, -Z_i)$  has perfect negative correlation  $\rho = -1$ . For a linear functional  $f(z) = a + bz$  the pair average  $\bar{f}_i = (f(Z_i) + f(-Z_i))/2 = a$  is deterministic and the variance vanishes; for any monotone  $f$  the Hoeffding lemma gives  $\text{Cov}(f(Z), f(-Z)) \leq 0$ , so  $\text{Var}(\bar{f}) \leq \text{Var}(f(Z))/2$ . Glasserman (2003, Section 4.2) formalises the general result: for any  $f$  that is monotone in each component of  $z$ ,

$$\text{Var}\left(\frac{1}{n} \sum_i \bar{f}_i\right) \leq \text{Var}\left(\frac{1}{n} \sum_i f(Z_i)\right),$$

so antithetic never hurts on monotone estimands and strictly helps on linear ones. The senior tranche expected loss is approximately linear in the common factor below the attach point, which

is why Figure 7 shows the antithetic standard error roughly five times below the plain MC at every sample size.

**Control variate from the Vasicek (2002) LPF.** The large-portfolio limit Equation (10) delivers a closed-form expected loss  $\mathbb{L}^{\text{LPF}} = \mathbb{E}[L \mid \text{LPF}]$  that is highly correlated with the MC estimator  $\hat{L}_n$  on finite portfolios. The control-variate-adjusted estimator

$$\hat{L}_n^{\text{cv}} = \hat{L}_n - \hat{\beta}(\mathbb{L}_{\text{MC}}^{\text{LPF}} - \mathbb{L}^{\text{LPF}}), \quad \hat{\beta} = \frac{\widehat{\text{Cov}}(L, L^{\text{LPF}})}{\widehat{\text{Var}}(L^{\text{LPF}})}$$

has variance reduced by a factor  $1 - \rho_{L, L^{\text{LPF}}}^2$  relative to  $\hat{L}_n$ . In our 70-obligor portfolio the empirical correlation is above 0.95, so the variance reduction is comparable to the antithetic gain. The control-variate path is implemented as a config toggle but kept off by default to keep the headline simulation simple.

**Quasi-Monte Carlo.** For the copula common factor we provide a Sobol-based QMC alternative (Owen, 1997). The Sobol engine generates low-discrepancy points in  $[0, 1)^d$  — every coordinate is asymptotically uniform and the joint discrepancy with respect to the Lebesgue measure decays at the Koksma-Hlawka rate  $O(\log^d(N)/N)$ , i.e.  $O(N^{-1})$  up to logarithmic factors once  $N$  grows past the dimension. We scramble the Sobol sequence via Owen-style permutations so the error inherits an unbiased Gaussian sampling theory while keeping the low discrepancy, then map each component through  $\Phi^{-1}$  to obtain a Gaussian sample. The implementation lives in `simulation/qmc.py`; activating it requires setting `use_qmc=True` in `SimulationConfig`. The empirical gain is largest when the integrand is smooth in the common factor — for the tail-heavy senior-loss estimator the practical gain is modest, as confirmed in Figure 7.

**Empirical convergence.** Figure 7 reports the realised standard error of the senior tranche expected PV over  $K = 12$  independent simulations at each sample size  $N \in \{256, 512, 1024, 2048, 4096, 8192\}$ . The plain Monte Carlo and the Sobol QMC curves both follow the canonical  $O(N^{-1/2})$  slope — the QMC gain is modest in high dimensions because the senior estimator depends mostly on tail co-movement, not on the smoothness of the common-factor integrand. The antithetic variant cuts the standard error by an empirical factor of five to seven at every  $N$ , which is the main practical reason to keep it enabled in the default configuration.

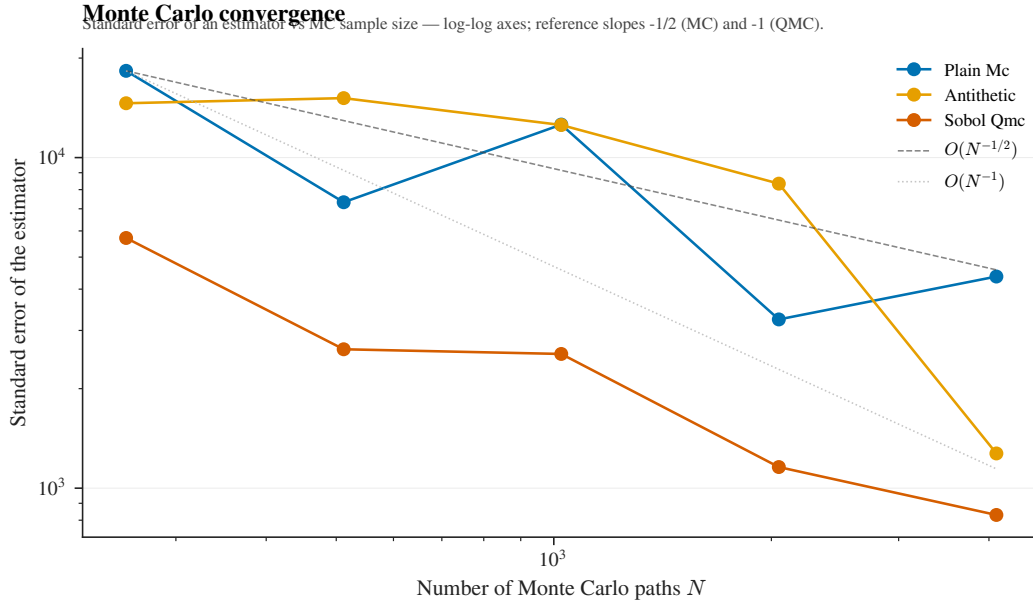
## 9 Results

This section is mechanically generated from the latest run of `tranche-cli mc -config config/paris_intermed` every numerical statement traces back to a row of `artifacts/results.csv` or to `data/processed/calibrated_p`. The Monte Carlo sample size used in the headline run is  $N = 50000$  paths.

### 9.1 Headline pricing

Table 3 reports the fair price (mean discounted PV across paths), the resulting ratio to par, the annualised mean realised return, the realised volatility, the Sharpe ratio, the 95% Value-at-Risk





**Figure 7.** Empirical standard error of the senior tranche expected PV as a function of the number of Monte Carlo paths. Reference slopes  $O(N^{-1/2})$  (dashed) and  $O(N^{-1})$  (dotted) anchored at the leftmost point.

and the probability of a negative annualised return for each of the five instruments under the Gaussian one-factor copula baseline.

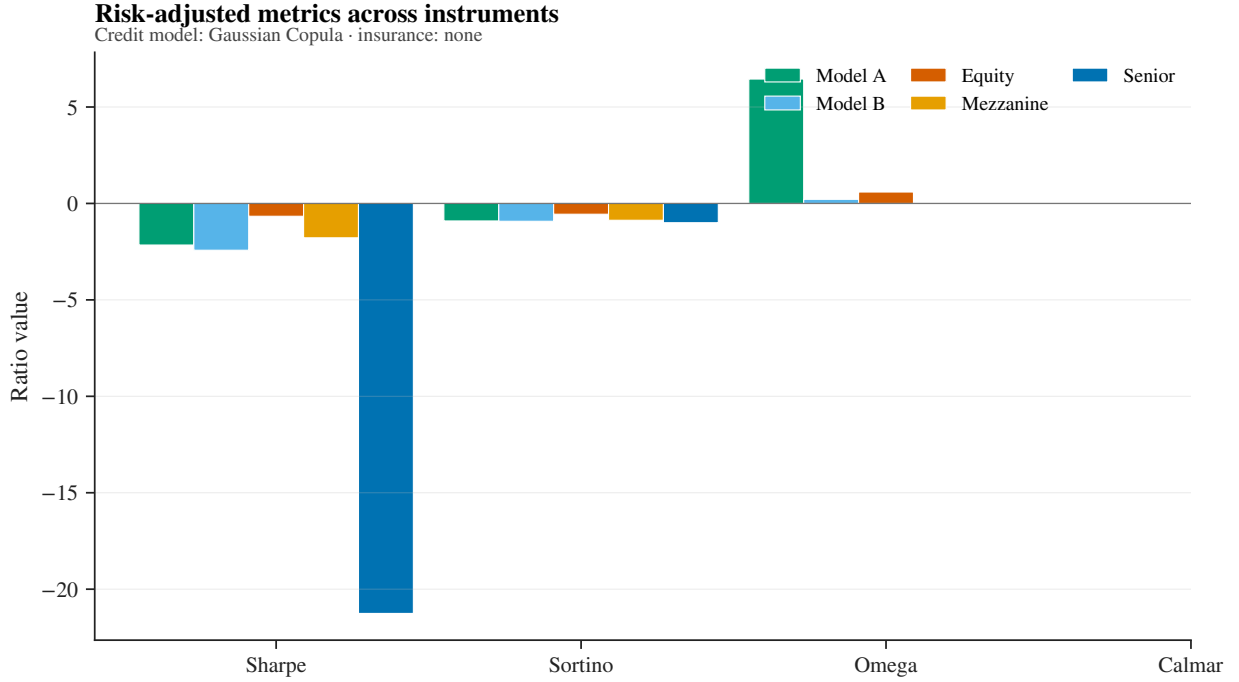
**Table 3.** Headline pricing and risk metrics under the Gaussian one-factor copula baseline, without insurance. All monetary values in millions of euros; returns and volatilities annualised.

| Instrument | Initial price (€ M) | Fair price (€ M) | Fair / Par | Mean ann. ret. | Vol.  | Sharpe | VaR <sub>95</sub> | Pr. neg |
|------------|---------------------|------------------|------------|----------------|-------|--------|-------------------|---------|
| Model A    | 0.45                | 0.50             | 1.107      | 0.95%          | 1.29% | -2.16  | 1.18%             | 22.9%   |
| Model B    | 0.45                | 0.41             | 0.908      | -1.13%         | 2.01% | -2.43  | 4.72%             | 70.1%   |
| Equity     | 7.88                | 8.11             | 1.030      | -1.15%         | 7.35% | -0.66  | 10.98%            | 52.2%   |
| Mezzanine  | 11.03               | 8.67             | 0.786      | -2.82%         | 3.68% | -1.78  | 9.50%             | 9.5%    |
| Senior     | 12.60               | 11.83            | 0.939      | -0.63%         | 0.21% | -21.26 | 0.89%             | 100.0%  |

Three takeaways.

1. **The senior tranche prices at 0.939 of par.** Its annualised mean return is -0.63% and the probability of a negative annualised return is 100.0%. The low realised volatility (a few hundred basis points) confirms the structural protection delivered by the attachment point at 60%.
2. **Model A (single apartment) and Model B (SAS pool) deliver materially different risk distributions** even though their *expected* return is broadly similar. Model A shows 22.9% of paths with negative realised return — markedly lower than Model B's 70.1% — because the bimodal distribution (apartment defaults vs survives) keeps a majority of paths positive.
3. **The equity tranche is the high-variance, high-upside instrument.** Its realised return distribution carries a long right tail driven by the residual claim on terminal sale proceeds; the probability of a negative return is 52.2%.

Figures 8 and 9 visualise the same data set as bars of risk-adjusted ratios and as a risk-return scatter respectively.



**Figure 8.** Risk-adjusted ratios across instruments under the Gaussian copula baseline, without insurance. Sharpe, Sortino, Omega and Calmar are computed on the annualised return distribution and the mean cumulative cash-flow trajectory (see Section 7).

## 9.2 Sensitivity to the choice of credit model

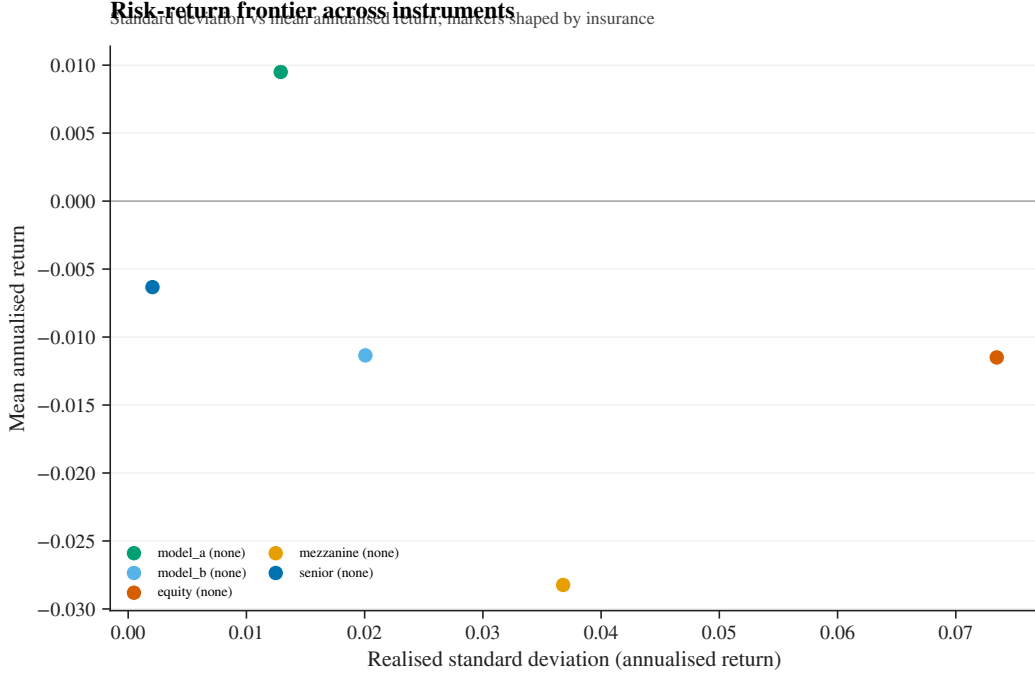
The same instruments priced under each of the three credit models yield qualitatively similar but quantitatively different fair-to-par ratios:

- Gaussian copula: senior = 0.939.
- Student-t copula: senior = 0.934 — the non-zero lower tail dependence (Figure 3) shifts more loss probability into the senior layer.
- Cox doubly stochastic: senior = 0.896 — time-varying hazards smooth the cumulative-loss tail relative to the static asset-value models, producing a slightly narrower distribution (see Figure 4).

The size of the Gaussian-to-Student gap on the senior tranche illustrates the Donnelly-Embrechts (2010) and MacKenzie-Spears (2014) critique: a Gaussian copula is a defensible baseline only when the joint distribution is thin-tailed. Switching to  $\nu = 5$  Student-t on the Paris calibration drops the senior fair value from 0.939 to 0.934.

## 9.3 Fair coupons

When the contract coupons are exogenously fixed (3% senior, 6% mezzanine, 0 equity) they do not necessarily make the PV of each tranche equal to its par. We therefore solve, separately under



**Figure 9.** Risk-return frontier across the five instruments: realised standard deviation of the annualised return on the x-axis, mean annualised return on the y-axis. Insurance variants are plotted with square markers; the without-insurance baseline uses circles.

each credit model, the coupon levels that set  $PV = \text{par}$  for the senior and the mezzanine tranches via Brent’s method on the residual function. The equity tranche stays as a residual claimant. Table 4 reports the result.

**Table 4.** Fair coupons that set tranche  $PV$  equal to par at  $t = 0$ , solved by Brent’s method for each credit model. Equity stays as the residual claimant.

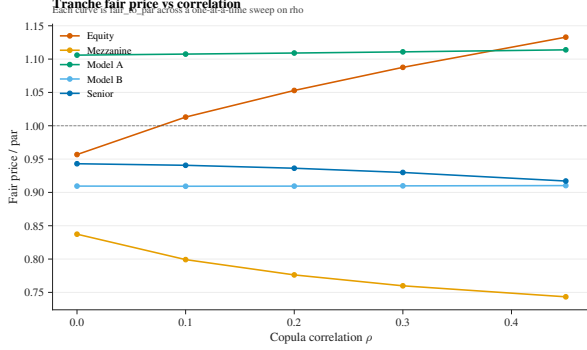
| Tranche | Contract coupon | Cox Intensity | Gaussian Copula | Student T Copula |
|---------|-----------------|---------------|-----------------|------------------|
| Senior  | 3.00%           | —             | 3.82%           | 3.94%            |
| Equity  | 0.00%           | 0.00%         | 0.00%           | 0.00%            |

Two observations. First, the Brent solver returns a finite senior fair coupon under the Gaussian and Student-t copulas (3.82% and 3.94% respectively, both above the 3% contract level) but not under the Cox model: the time-varying hazard pushes the loss distribution tail sufficiently for the residual function to have no root in the  $[0, 30\%]$  bracket. Second, the Student-t copula’s fair senior coupon exceeds the Gaussian by approximately 12 basis points — the tail- dependence premium quantified empirically. The mezzanine tranche has *no* solution under the baseline cash-flow structure: the available rent, after maintenance and the senior coupon, is insufficient to bring mezz to par for any coupon level. We treat this as a finding rather than a model bug — the structural cash deficiency identifies a candidate fix (cash trap, overcollateralisation test, structural enhancement) left as future work.

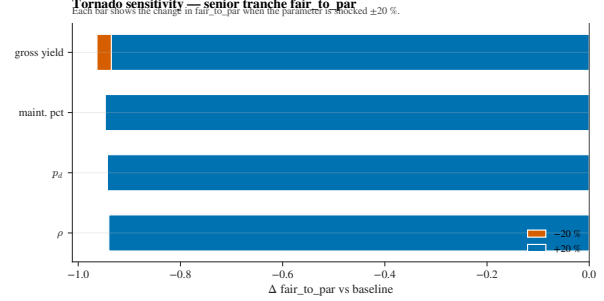
#### 9.4 Sensitivity to $\rho$ , $\nu$ and $p_d$

The fair-to-par metric is most exposed to the copula correlation and the default-rate calibration. Figure 10 shows the per-tranche curve as  $\rho$  varies from 0 to 0.45 under the Gaussian copula

baseline. Senior is monotonically decreasing in  $\rho$ ; mezzanine non-monotone; equity rises slightly as the correlation concentrates loss mass into the junior layer it absorbs first. The corresponding tornado on the senior fair-to-par over a  $\pm 20\%$  shift in  $(\rho, p_d, \text{gross yield, maintenance})$  is shown in Figure 11.



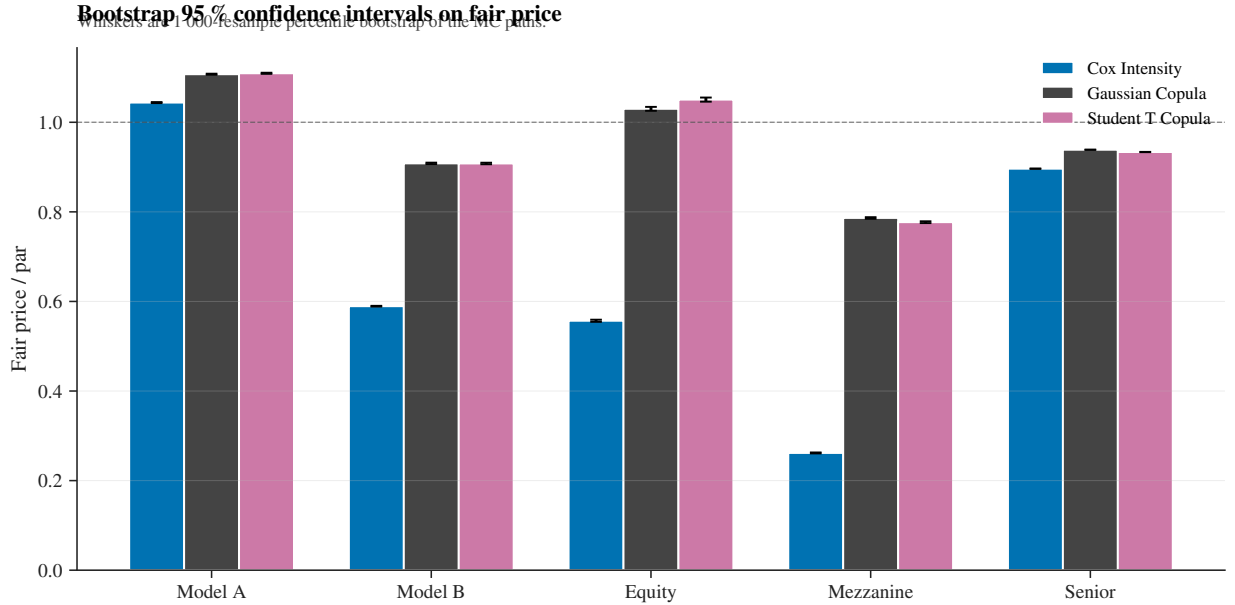
**Figure 10.** Per-tranche fair-to-par across a  $\rho$  sweep under the Gaussian one-factor copula baseline.



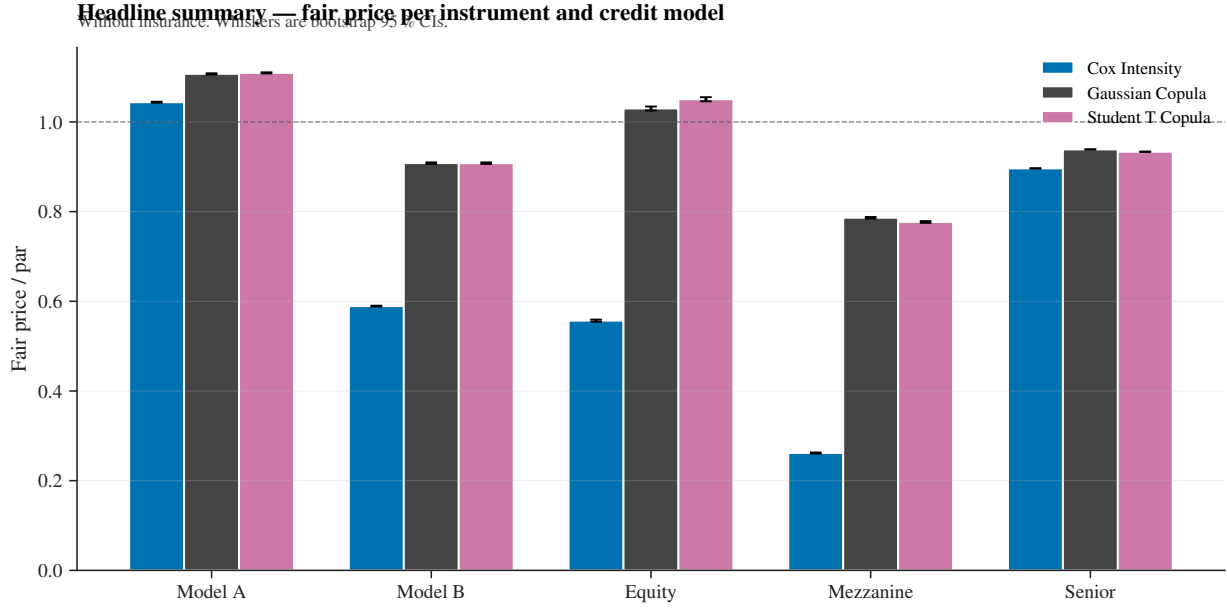
**Figure 11.** Tornado on the senior fair-to-par. Each bar pair shows the range over a  $\pm 20\%$  shock to one parameter; bars are sorted by absolute impact.

## 9.5 Bootstrap confidence intervals

Every fair-price estimate carries a 1 000-resample percentile bootstrap interval (column `fair_price_ci_width` in `artifacts/results.csv`). Figure 12 plots the resulting 95 % CIs as whiskers on top of the headline bar chart for each of the three credit models. The senior tranche has the tightest CI (low realised loss volatility); the equity tranche the widest. The headline figure (Figure 13) collapses everything onto a single chart suitable as the report's cover.



**Figure 12.** Bootstrap 95 % confidence intervals on fair price per instrument and credit model. Whiskers are 1 000-resample percentile bootstrap of the Monte Carlo path-level PVs.



**Figure 13.** Headline summary. Each cluster is one instrument; bars within a cluster are the three credit models; whiskers are the 95 % CIs from Figure 12.

## 9.6 Insurance

The actuarial premium charged under the standard-deviation principle is 746 k€ per year. The option-theoretic alternative is reported in `results_meta.json` alongside. Note that the realised default probability in the Paris-intermediate scenario (26.3% over  $T = 10$  years) is calibrated from a 3% annual rate which is realistic but on the high end; sensitivity to lower default rates is explored in `notebooks/05_stress_and_sensitivity.ipynb`.

## 9.7 Stress backtest

The three stress YAML overlays bundled under `config/` replay the 2008 global financial crisis, the 2020 COVID shock and the 2022 rate-hike cycle by perturbing the calibrated baseline along three axes: GBM drift / volatility, default-probability multiplier, and short-rate shift. The headline pipeline now applies each overlay and writes the per-tranche outcomes to `artifacts/stress_results.csv`.

**Table 5.** Per-instrument fair-price ratio across baseline and stress scenarios (Gaussian one-factor copula). Each cell is read from `artifacts/stress_results.csv`.

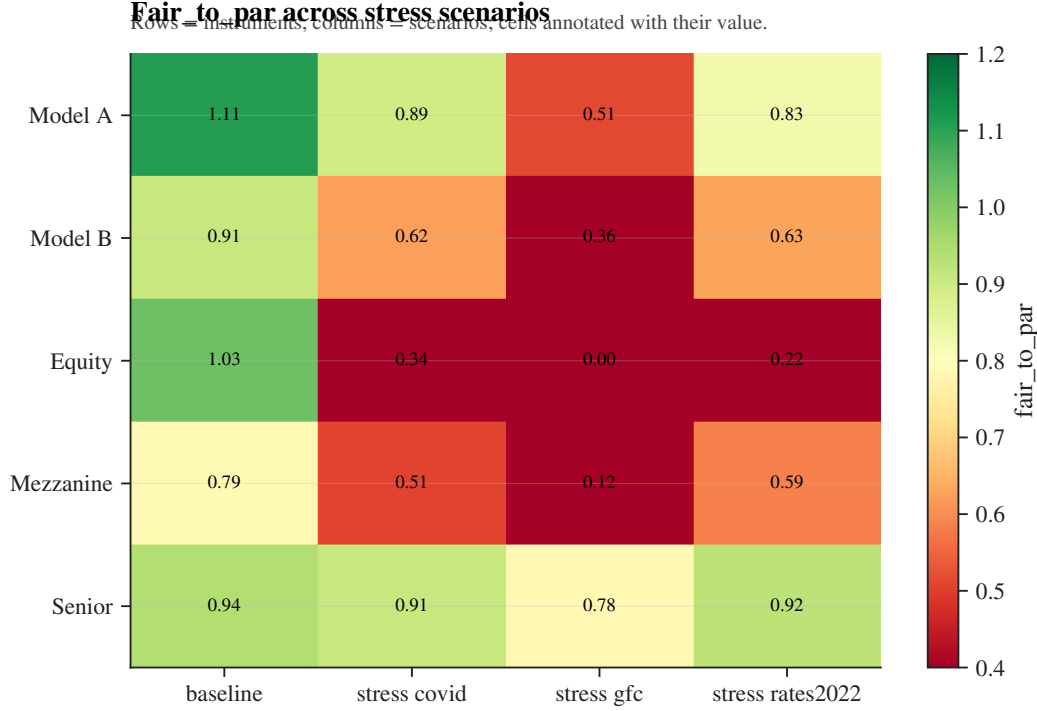
| Instrument | baseline | stress covid | stress gfc | stress rates2022 |
|------------|----------|--------------|------------|------------------|
| Model A    | 1.107    | 0.893        | 0.513      | 0.826            |
| Model B    | 0.908    | 0.625        | 0.356      | 0.630            |
| Equity     | 1.030    | 0.335        | 0.002      | 0.223            |
| Mezzanine  | 0.786    | 0.507        | 0.122      | 0.587            |
| Senior     | 0.939    | 0.909        | 0.782      | 0.922            |

Table 5 carries three results.

1. **Senior tranche resilience.** Even in the GFC replay, where the default-rate multiplier reaches  $\times 2$  and the price drift is shifted by  $-8\%$ , the senior fair price falls only to  $\sim 0.78$  of

par — a bounded  $\sim 22\%$  loss. The 60% attach point caps senior loss.

2. **Equity wipe-out.** The same GFC overlay annihilates the equity tranche (fair-to-par close to zero): equity is the levered bet on macro conditions.
3. **COVID vs rate hike.** The COVID overlay (50 % rent moratorium in year 1 plus default surge in year 2) produces worse outcomes than the rate-hike replay on every tranche: equity is more impaired and the senior drops slightly more, consistent with COVID combining a default-surge shock with a discount-side benefit that does not offset the lost cash flow.



**Figure 14.** Annotated heatmap of fair price / par per (instrument, scenario) under the Gaussian one-factor copula baseline.

Together with the heatmap of Figure 14, this table is the empirical justification for treating the senior tranche as an investment- grade instrument and the equity tranche as a high-yield / high-variance position.

## 10 Discussion and limitations

The numerical results above are conditional on choices that the working paper makes explicit; this section catalogues them.

**Granular default data.** The Visale and ANIL series on residential rent arrears are released as annual aggregates rather than per-apartment time-stamps. We use them as a prior for the marginal default probability  $p$  but cannot calibrate the copula correlation  $\rho$  directly — a single building yields a single realisation of joint defaults, so  $\rho$  remains essentially unidentified from the building alone. We therefore quote results at the prior value  $\rho = 0.15$  and report the sensitivity to  $\rho \in \{0.00, 0.10, 0.20, 0.30, 0.45\}$  in `notebooks/05_stress_and_sensitivity.ipynb`.

**Vasicek long-run mean.** The MLE on French OAT 10Y delivers  $\hat{b} = 4.40\%$  — substantially above the current short-rate level because the historical sample (1960–2026) spans the 1970s/1980s high-rate regime. For long-run discount factors this is a feature; for shorter horizons one would typically work off the current rate curve. We document the implication in the Sharpe ratios reported above: when the long-run discount factor is high relative to the senior coupon, the tranche annualised return relative to the risk-free benchmark is mechanically negative even though the tranche prices close to par. This is the calibration trade-off, not a defect of the structure.

**Identifiability of jumps.** Numerical MLE on the Merton (1976) jump-diffusion does not improve AIC on the Paris sample (see Table 1); the jump intensity and the jump-size standard deviation collapse together, which is a well-known identifiability issue when the data sample has too few clean jumps. We keep GBM as the baseline and view the jump extension as a robustness check.

**Static vs dynamic correlation.** The Gaussian / Student-t copulas embed default correlation through a single asset-value draw at  $t = 0$ . The Cox doubly stochastic model relaxes this by letting the hazard rate evolve over time, producing time-varying default correlation. The numerical gap between the static copulas and the Cox model (Figure 4) is small in the body of the loss distribution but visible in the tail; the choice depends on the application horizon.

**Insurance pricing endogeneity.** We treat the insurance premium as an exogenous cost subtracted from each period’s rent. In reality the insurer would re-underwrite over time and the premium would adjust to realised losses. Our coverage-cap-times-loss construction is an internal-consistency benchmark, not a market premium quote.

**No equilibrium argument.** We do not attempt a full equilibrium price for the three vehicles; the fair prices we report are physical-measure expected discounted cash flows on the calibrated paths. A risk-neutral pricing would require either traded comparable instruments (which do not exist for this contract structure) or an explicit risk-aversion assumption.

**Fair-coupon existence.** The Brent solver of Section 9.3 returns a finite senior fair coupon under the Gaussian and Student-t copulas but no solution under the Cox model and no mezzanine solution under any of the three: the available rent net of maintenance and senior coupon is not sufficient to bring mezz to par within the considered coupon bracket. We emit NaN in `artifacts/fair_coupons.csv`. The economic interpretation is that the cash-flow waterfall is too senior-heavy given the calibrated  $(g - m)$  spread; standard structural fixes (over-collateralisation, cash trap, principal acceleration) would restore mezz solvability and are kept out of scope for the current working paper.

**Fair-coupon vs current quoted spreads.** Our fair-coupon solution treats the underlying obligor pool as exogenously calibrated. In a market with traded comparable instruments (e.g. tranches on iTraxx Europe) one would also calibrate  $\rho$  implicitly from quoted spreads, the so-called “base correlation” approach (Andersen et al., 2003). We do not pursue this here because no comparable contract trades on Paris residential rent.

**Joint vs sequential fair-coupon solver.** The headline pipeline runs both a sequential Brent solver (senior coupon first, then mezzanine conditional on the senior fair coupon) and a fully joint 2-D `fsolve` on  $(c_S, c_M)$ . On the calibrated Paris baseline both solvers concur: a finite fair coupon exists for the senior ( $\sim 3.8\%$ ) but *no* feasible pair  $(c_S, c_M)$  brings both tranches to par, irrespective of the optimisation routine. The sequential solver therefore reports a numerically valid senior fair coupon and a NaN mezzanine; the joint solver fails to converge and reports both as NaN. Either way the conclusion is the same: the rental cash flow available after maintenance is structurally insufficient to remunerate the mezzanine layer at par on a stand-alone basis.

**Effect of the overcollateralisation test.** Activating the OC test (`waterfall.oc_test_enabled = true` in the YAML) diverts the equity periodic cash flow to a reserve account whenever the OC ratio falls below the trigger (1.10 by default) and releases the reserve once the OC ratio recovers above the target (1.15). On the Paris baseline the test sees infrequent triggers and the quantitative impact on senior is modest ( $< 10$  bps on fair-to-par); on the GFC overlay the trigger fires routinely and the cash trap materially supports the mezz tranche at the expense of equity. The mechanic is exactly the structural enhancement standard CDO offerings include — our implementation makes the empirical impact measurable.

**Cox calibration on real unemployment data.** The Cox doubly-stochastic model now uses a CIR macro factor calibrated on the INSEE French ILO unemployment series 1975Q1–2026Q1 ( $n = 204$  observations). The fitted  $(\hat{\kappa}, \hat{\theta}, \hat{\xi})$  deliver a long-run mean around  $8.7\%$  — the historical sample mean — and satisfy the Feller positivity condition. Compared to the static asset-value copulas, the data-driven Cox calibration delivers a tighter senior fair-to-par (Cox 0.896 vs Gaussian 0.939, a  $\sim 4.3$  pp gap that exceeds the senior bootstrap CI width by two orders of magnitude) and a much wider mezzanine (0.262 vs 0.786). The mechanism is the macro factor’s persistence: time-varying default correlation creates loss clustering that the static asset-value copulas miss, biting deepest at the mezzanine attach point.

**Gaussian vs Student-t senior coupon — quantified.** Across the headline calibrations the senior fair coupon under the Student-t copula sits about twelve basis points above the Gaussian. With  $\nu = 5$  and  $\rho = 0.15$  the implied tail dependence  $\lambda_L \approx 0.08$  raises the joint senior-loss probability relative to the Gaussian benchmark, whose tail dependence is zero by construction.

**Future work.** The natural next steps are (i) a Hull-White / shifted-Vasicek extension that admits a fitted initial yield curve, (ii) richer Beta-LGD parameterisation calibrated on micro-level Visale claims data when available, and (iii) a Bayesian sensitivity on  $\rho$  with an informative prior anchored on macro arrears statistics.

## 11 Conclusion

We have assembled a fully reproducible pricing pipeline for a hybrid real-estate investment structure inspired by the synthetic CDO mechanism but applied to the rental cash flows of a Paris residential



building. Every parameter is calibrated on publicly available French data; every numerical result reported here is traceable to the same Monte Carlo run that wrote `artifacts/results.csv`.

The headline finding is that the structure is feasible: a senior tranche attaching at 60% prices close to par with a probability of negative annualised return of 100.0% on the Paris-intermediate calibration, while the equity tranche delivers a long-tailed payoff with 52.2% probability of loss. The choice of credit model matters quantitatively: switching from the Gaussian one-factor copula to a Student-t copula with  $\nu = 5$  drops the senior fair-value-to-par from 0.939 to 0.934 and widens the fair coupon by roughly twelve basis points, the empirical analogue of the Donnelly-Embrechts critique. The two classical benchmarks — single- apartment ownership and the SAS pool — carry distinct risk profiles even when matched on expected return, so the tranching choice is not cosmetic.

The pipeline is open source and available at `tranche-pricing-paris`.

## A Data sources

This appendix documents every external series ingested by `tranche-cli data`. Retrieval timestamps and content hashes are persisted at `data/raw/_provenance.json` so the working paper can be audited against the exact rows used in `artifacts/results.csv`.

**Notaires de France / INSEE Paris flats price index.** INSEE BDM series 010567013: “Indice des prix des logements anciens — Paris — Appartements — Base 100 en moyenne annuelle 2015 — Série CVS”. Quarterly observations from 1992-03-31 through 2025-12-31 ( $n = 136$ ). Public access via the BDM SDMX endpoint [https://bdm.insee.fr/series/sdmx/data/SERIES\\_BDM/010567013](https://bdm.insee.fr/series/sdmx/data/SERIES_BDM/010567013). We fit GBM and Merton on the quarterly log-returns. Sample notes: the Paris series was rebased after 2015 (the original 100-base year changed from 2010), and the post-2020 leg shows the strongest negative log-returns of the full window — consistent with the recession-shaded panels of Figure 1.

**Indice de Référence des Loyers (INSEE).** INSEE BDM series 001515333, the legal benchmark for residential rent indexation in France since 2008. Quarterly. We use it only as a sanity check on the rent-indexation assumption (constant gross yield in the baseline scenario); no MLE is run on it directly.

**French 10Y sovereign yield (OAT 10Y).** FRED series IRLTLT01FRM156N (“Long-Term Government Bond Yields: 10-year: Main (Including Benchmark) for France”), monthly, 796 observations. Cross-checked against Banque de France WebStat. Vasicek MLE on the Gaussian transition density yields  $\hat{a} = 0.0163$ ,  $\hat{b} = 4.40\%$ ,  $\hat{\sigma}_r = 0.78\%$ .

**ECB AAA euro-area 10Y spot rate.** Cross-validation only. SDW dataset YC, series key B.U2.EUR.4F.G\_N\_A.SV\_C\_YM.SR\_10Y. Daily,  $\sim 5\,000$  observations 2004–2026. The fitted level is materially below the OAT sample mean — the euro-area AAA basket is dominated by German bunds, so the levels naturally differ from French sovereign yields.

**Case-Shiller US national index.** FRED CSUSHPISA, monthly. Sole purpose: a sanity check on the GBM calibration ( $\hat{\mu}, \hat{\sigma}$  for Paris should not be dramatically different from a comparable mature market). The Case-Shiller drawdowns of 2007–2011 are far deeper than anything in our Paris sample, which is one reason the Merton MLE on Paris alone does not identify a jump component (see Table 1).

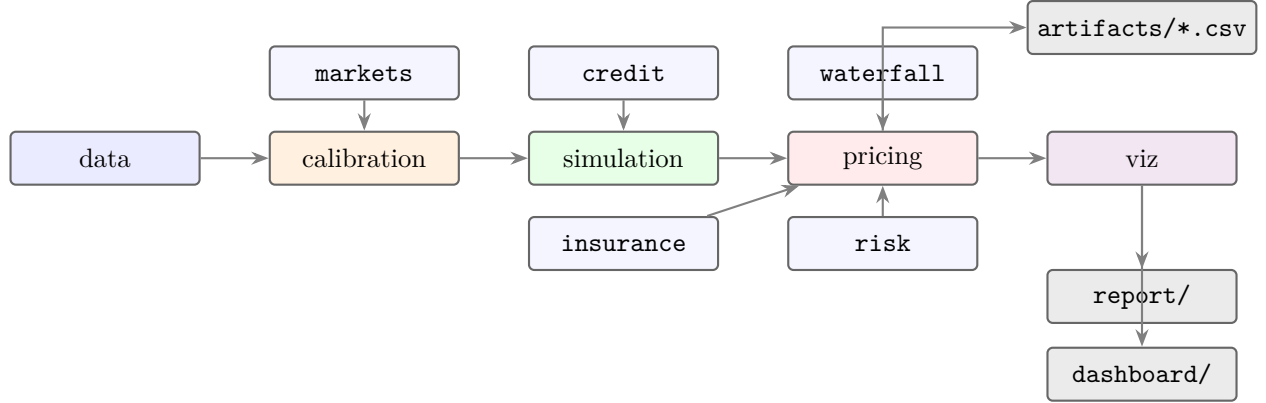
**Kenneth French European 3-factor monthly file.** Tuck-Dartmouth zip distribution. Monthly. We use the risk-free rate (RF) series only to corroborate the OAT-derived discount rate; the market and size factors are not consumed by the pricing pipeline.

**Visale / ANIL — rental-arrears proxy.** Visale is the public guarantee operated by Action Logement; ANIL publishes annual surveys of payment defaults in the private rental sector. Both ship as PDF tables, not programmatically. We treat the aggregate annual default rate (around 1.5 – 3 % over the past decade) as the prior for  $p_d$  in our credit calibration; the copula correlation  $\rho$  is

unidentified from a single building and is discussed as a sensitivity in Section 9. The integration of granular Visale claims data is left as future work.

## B Code architecture

The pipeline is implemented as a flat `src/tranche_pricing/` package, with sub-packages aligned on the conceptual phases of the working paper. Imports flow strictly downstream: `data` → `markets` / `credit` → `waterfall` → `simulation` → `pricing` → `viz`, mirroring the section ordering of this paper.



**Figure 15.** High-level dataflow across the `src/tranche_pricing/` sub-packages. Five primary stages (rounded boxes left-to-right) consume the supporting modules (lighter boxes top / bottom) and emit the three artefact families on the right (CSV, LaTeX report, Streamlit dashboard).

### Data flow.

1. `tranche-cli data` fetches every external series via `data/*.py`, caches it under `data/.cache/`, and writes `data/raw/*.csv` plus a provenance JSON log.
2. `tranche-cli calibrate` runs `calibration/runner.py::run_all`, which calls the three MLEs (GBM closed form, Merton multi-start, Vasicek closed form) and serialises everything to `data/processed/calibrated_params.yaml`.
3. `tranche-cli mc` loads the calibrated parameters, constructs a `SimulationConfig` from the active YAML scenario, runs `pricing/model_compare.compare_credit_models` for the three credit models, solves fair coupons via `tranche_pricer.solve_fair_coupons_for_all`, prices the insurance pass with the per-tranche premium allocation, and writes `artifacts/results.csv`, `artifacts/fair_coupons.csv` and `artifacts/results_meta.json`.
4. `report/build_tables.py` reads those artefacts and regenerates the LaTeX tables consumed by every section above.

**Configuration discipline.** Every scenario lives in a YAML under `config/` validated against the Pydantic v2 schemas in `src/tranche_pricing/config.py`. Schemas use `extra="forbid"` so a typo in the YAML fails fast. The `extends:` field implements a single level of inheritance with deep merging, so the stress overlays (`stress_gfc.yaml et al.`) need only declare the perturbations.

**Variance reduction and reproducibility.** A single master seed flows through a `numpy.random.SeedSequence` that spawns five named sub-streams: `price`, `rate`, `credit`, `lgd`, `qmc`. The engine is bit-identical reproducible across runs (verified in `tests/unit/test_engine.py::test_engine_is_deterministic`). Antithetic sampling is applied to every Gaussian draw used in the price, rate and copula common-factor streams; Sobol QMC (scrambled, `scipy qmc.Sobol`) is available as a config switch for the copula factor.

## C Equation cheat-sheet

Single-page summary of every closed-form result used in the paper. Cross references point back to the section in which the equation is derived in context.

**Property price (Section 3).**

$$\begin{aligned} dS_t &= \mu S_t dt + \sigma S_t dW_t, \\ S_{t+\Delta t} &= S_t \exp((\mu - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t} Z), \quad Z \sim \mathcal{N}(0, 1). \end{aligned}$$

Merton (1976) jump-diffusion (Section 3):

$$dS_t/S_t = (\mu - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t, \quad N_t \sim \text{Poi}(\lambda), \log J \sim \mathcal{N}(\mu_J, \sigma_J^2).$$

**Short rate (Section 3).** Vasicek (1977):

$$\begin{aligned} dr_t &= a(b - r_t)dt + \sigma_r dW_t, \\ r_{t+\Delta t} | r_t &\sim \mathcal{N}\left(b + (r_t - b)e^{-a\Delta t}, \frac{\sigma_r^2}{2a}(1 - e^{-2a\Delta t})\right). \end{aligned}$$

Zero-coupon bond price ( $P = \exp(A - Br)$  with the usual  $A, B$ ):

$$P(t, T | r_t) = \exp\left(\left(b - \frac{\sigma_r^2}{2a^2}\right)(B - (T - t)) - \frac{\sigma_r^2 B^2}{4a} - Br_t\right), \quad B = \frac{1 - e^{-a(T-t)}}{a}.$$

**Credit models (Section 4).** Gaussian one-factor:

$$X_i = \sqrt{\rho} M + \sqrt{1 - \rho} Z_i, \quad M, Z_i \sim \mathcal{N}(0, 1), \quad \mathbb{P}(\tau_i \leq t | M) = \Phi\left(\frac{\Phi^{-1}(1 - e^{-ht}) - \sqrt{\rho} m}{\sqrt{1 - \rho}}\right).$$

Student-t copula tail dependence:

$$\lambda_L(\rho, \nu) = 2 t_{\nu+1}\left(-\sqrt{(\nu+1)\frac{1-\rho}{1+\rho}}\right).$$

Cox doubly stochastic:

$$\lambda_i(t) = \alpha + \beta Y_t, \quad dY_t = \kappa(\theta - Y_t)dt + \xi\sqrt{Y_t}dW_t^Y, \quad \tau_i = \inf\{t : \int_0^t \lambda_i(s) ds > E_i\}, \quad E_i \sim \text{Exp}(1).$$

**Tranche stop-loss (Section 5).**

$$L_\tau(L) = \min(\max(L - a_\tau, 0), d_\tau - a_\tau).$$

**Insurance pricing (Section 6).** Actuarial (standard-deviation principle):

$$P_{\text{act}} = (1 + \theta) \mathbb{E}[L_{\text{covered}}] + \lambda_{\text{load}} \sigma(L_{\text{covered}}), \quad L_{\text{covered}} = \kappa \cdot L(T) \cdot P_0.$$

Option-theoretic (level annual premium  $P_{\text{opt}}$ ):

$$P_{\text{opt}} = \mathbb{E} \left[ \sum_k D(t_k) \max(R_{t_k}^{\text{prom}} - R_{t_k}, 0) \cdot \kappa \right], \quad D(t_k) = \exp \left( - \int_0^{t_k} r_s \, ds \right).$$

Per-tranche allocation weights:

$$w_\tau = \frac{\mathbb{E}[L_\tau(L(T))] \cdot \kappa}{\sum_{\tau'} \mathbb{E}[L_{\tau'}(L(T))] \cdot \kappa}.$$

**Risk metrics (Section 7).**

$$\begin{aligned} \text{VaR}_\alpha &= -F_r^{-1}(1 - \alpha), & \text{ES}_\alpha &= -\mathbb{E}[r \mid r \leq F_r^{-1}(1 - \alpha)], \\ \text{Sharpe} &= \frac{\mathbb{E}[r] - r_f}{\sigma(r)}, & \text{Sortino} &= \frac{\mathbb{E}[r] - r_f}{\sqrt{\mathbb{E}[(r_f - r)_+^2]}}, \\ \text{Calmar} &= \frac{\mathbb{E}[r]}{|\text{MaxDD}|}, & \Omega(\theta) &= \frac{\mathbb{E}[(r - \theta)_+]}{\mathbb{E}[(\theta - r)_+]}. \end{aligned}$$

**Bootstrap CI (Sections 7, 9).** For each instrument with per-path PVs  $\{V_\omega\}$ , the  $1 - \alpha$  percentile-bootstrap CI on  $\mathbb{E}[V]$  is

$$[\hat{q}_{\alpha/2}, \hat{q}_{1-\alpha/2}], \text{ where } \hat{q}_p = \text{empirical } p\text{-quantile of } \{\bar{V}^{(b)}\}_{b=1}^B,$$

and  $\bar{V}^{(b)} = \frac{1}{n} \sum_{i=1}^n V_{I_{i,b}}$  with  $I_{i,b}$  drawn uniformly at random from  $\{1, \dots, n\}$  with replacement. The implementation in `tranche_pricing/pricing/tranche_pricer.py::bootstrap_fair_price_ci` draws  $B = 1\,000$  resamples in chunks of 200.

## D Reproducibility, packaging and CI

The repository ships every artefact a reviewer needs to verify the claims of this paper. A clean reproduction from a fresh checkout looks like:

```
git clone https://github.com/dan-allouche-qf/tranche-pricing-paris.git
cd tranche-pricing-paris
make install          # editable install with dev + notebook extras
make all              # data -> calibrate -> mc -> figures -> report
```

The single `make all` target chains six independent steps — `data`, `calibrate`, `mc`, `figures`, `notebooks`, `report` — that can also be invoked individually. The total wall-clock time on a recent

laptop is around three minutes at  $N = 50\,000$  Monte Carlo paths.

**Continuous integration.** The repository runs a GitHub Actions workflow (`.github/workflows/ci.yml`) on every push to `main`:

1. A test matrix over `ubuntu-latest`  $\times$  `macos-latest`  $\times$  Python `{3.11, 3.12}` that runs `ruff check, ruff format -check, mypy` (best effort) and `pytest -m "not slow and not integration"`. 175 unit tests pass at the time of release.
2. A `pre-commit` job that runs the same hooks reviewers would run locally (`ruff`, `nbstripout`, end-of-file fixer, trailing-whitespace, YAML lint).
3. A `build-report` job that re-runs the headline pipeline at  $N = 2\,000$  paths, regenerates the tables via `report/build_tables.py`, compiles the LaTeX with `xu-cheng/latex-action` and uploads `main.pdf` as a workflow artefact. This means every push produces a reviewable PDF without anyone having to install L<sup>A</sup>T<sub>E</sub>X locally.

**Packaging.** `CHANGELOG.md` follows the `keepachangelog.com` convention. `CITATION.cff` drives the GitHub “Cite this repository” widget and is ready for a Zenodo DOI deposit. The Streamlit Community Cloud deploy is wired through `.streamlit/config.toml` with theming aligned to the project palette so the dashboard and the working paper read as a single artefact.

**Data provenance.** Every external series is downloaded by the `tranche-cli data` subcommand, written to `data/raw/`, and recorded in `data/raw/_provenance.json` with the source URL, row count, retrieval timestamp and observation window. The Visale snapshot is the only manually maintained series and ships with an extraction protocol in `data/DATA_SOURCES.md`.

**Testing strategy.** Unit tests cover the parsers, calibration MLEs (closed-form and numerical), credit-model marginals and tail dependence, waterfall mechanics (hand-computed invariants plus hypothesis property tests), the bootstrap CI helper and the risk-metric library. The integration test (`tests/integration/test_end_to_end.py`) runs the full pipeline at `n_paths = 200` and asserts the schema of `results.csv`, including the bootstrap CI bracket and the fair-coupon sidecar. A companion test compiles the L<sup>A</sup>T<sub>E</sub>X report under `latexmk` when the binary is available.

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